SEAT NO. SE-24009

NED UNIVERSITY OF ENGINEERING & TECHNOLOGY

FIRST YEAR (Software)

FALL SEMESTER EXAMINATIONS 2024

Time : 3 Hours | Batch 2024 | Dated : 06-DEC-24

Max Marks : 60

Discrete Structures - CT-162

Instructions:

- Attempt all questions
- Don't write the answers with pencils
- Neat and clean work shall attract more marks
- Attempt all parts of the same question at one place

Question#1 **(CLO-1)**

a) Let r and s be the propositions:
r: You go to the gym regularly.
s: You improve your health.

Find the propositions using r and s and logical connectives (including negations). **(6 marks)**

i) You do not go to the gym regularly. $\neg r$
ii) You go to the gym regularly, but you do not improve your health. $r \wedge \neg s$
iii) You will improve your health if you go to the gym regularly. $r \rightarrow s$
iv) If you do not go to the gym regularly, then you will not improve your health. $\neg r \rightarrow \neg s$
v) You improve your health, but you do not go to the gym regularly. $s \wedge \neg r$
vi) Whenever you improve your health, you are going to the gym regularly. $s \rightarrow r$

b) **Determine** whether $(\neg p \wedge (p \rightarrow q)) \rightarrow \neg q$ is a tautology. **(3 marks)** not

c) **Show** that $\neg p \rightarrow (q \rightarrow r)$ and $q \rightarrow (p \vee r)$ are logically equivalent using laws of logic. **(3 marks)**

Question#2 **(CLO-2)**

a) Let P(x) be the statement "x is a member of the basketball team" and let Q(x) be the statement "x is a member of the chess club."
Express each of these sentences in terms of P(x), Q(x), quantifiers, and logical connectives. The domain for quantifiers consists of all students at your school. **(8 marks)**

i. There is a student at your school who is a member of the basketball team and who is also a member of the chess club.
ii. There is a student at your school who is a member of the basketball team but who isn't a member of the chess club. ~~Every student at your school either can speak Russian or knows C++.~~
iii. Every student at your school is either a member of the basketball team or a member of the chess club.
iv. No student at your school is a member of the basketball team or a member of the chess club.

b) Determine the negations of the statements $\forall x(x2 > x)$ and $\exists x(x2 = 2)$? **(4 marks)**

x^2

i) $\exists x\ P(x) \wedge Q(x)$

ii) $\exists x\ P(x) \wedge \neg Q(x)$

iii) $\forall x\ P(x) \vee Q(x)$

iv) $\neg \forall x\ P(x) \vee Q(x)$

1 > 1
0 > 0
4 > 1

Question#3 **(CLO-2)**

a) Use set builder notation and logical equivalences to **show** $\overline{A \cap B} = \overline{A} \cup \overline{B}$, if A and B are sets. **(3 marks)**

b) Use a membership table to **show** that $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$, if A and B are sets. **(2 marks)**

c) Let A = {a, b, c, d, e} and B = {a, b, c, d, e, f, g, h}. Find
i) $A \cup B$. ii) $A \cap B$. iii) $A - B$. iv) $B - A$. **(4 marks)**

d) **Determine** the Cartesian product $A \times B$, where A is the set of courses offered by the mathematics department at a university and B is the set of mathematics professors at this university? Give an example of how this Cartesian product can be used. **(3 marks)**

Question#4 **(CLO-3)**

a) Let f be the function from {a, b, c, d} to {1, 2, 3, 4} with f(a) = 4, f(b) = 2, f(c) = 1, and f(d) = 3. **Determine** whether f is a bijection? **(4 marks)**

b) **Determine** whether the functions $f(x) = x + 1$ from the set of integers to the set of integers and $f(x) = x^2$ from the set of integers to the set of integers are onto? **(4 marks)**

c) **Solve** $(110101)_2 - (100101)_2$ using 1's compliment. **(2 marks)** 010000

d) **Calculate** the decimal value of 111001_2. **(2 marks)**

Question#5 **(CLO-3)**

a) **Apply** appropriate counting principle (Addition, product, subtraction and division rule to solve the following)

i) Six different airlines fly from New York to Denver and seven fly from Denver to San Francisco. How many different pairs of airlines can you choose on which to book a trip from New York to San Francisco via Denver, when you pick an airline for the flight to Denver and an airline for the continuation flight to San Francisco? **(2 marks)**

ii) A technology firm receives 500 applications from college graduates for a role in developing a new line of AI systems. Among these applicants: 300 majored in data science (~~including those with double majors~~), 200 majored in statistics (~~including those with double majors~~), and 75 majored in both data science and statistics. How many applicants majored neither in data science nor in statistics? **(2 marks)**

b) **Determine** how many students must be in a class to guarantee that at least two students receive the same score on the final exam, if the exam is graded on a scale from 0 to 50 points? **(2 marks)**

c) **Determine** the minimum number of students required in a biology class to be sure that at least four students will receive the same grade, if there are three possible grades: Excellent, Good, and Needs Improvement? **(2 marks)** 10

d) **Calculate** the value of **k** after the following code, (where n1, n2, … , nm are positive integers) has been executed? **(4 marks)**

```
k := 0
for i_1 := 1 to n_1
    k := k + 1
for i_2 := 1 to n_2
    k := k + 1
   .
   .
   .
for i_m := 1 to n_m
    k := k + 1
```

K = n1, n2, ... nm

300+200−75 = 425

500−425 = 75

6+7 = 13